MATH INQUIRERS HOLIDAY PROGRAM

Math Inquirers Question Bank

Primary 1

Notice it. Test it. Explain why.

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How to use this bank

Each lesson has three levels. Every child may start with Core, and any child may choose Think harder or Go further at any time. Nobody is placed in a level. Go further questions are good ones to talk about together.

CORE questions are low floor: a child who has not yet learnt the topic can start them with objects, a drawing, counting or guess-and-check. No special method is needed. Use them for warm-ups and for a confident start for every child.

THINK HARDER questions need the idea taught in the lesson, used in a new setting: two-step problems, patterns with a hidden rule, simple unknowns and larger numbers.

GO FURTHER questions are for any child who chooses them after the topic. They need the lesson idea plus an extension: more than one step, finding every way, working backwards, justifying an answer, changing one thing and asking what happens, or making up a puzzle of their own. Open questions have no single right answer; check the reasoning. This uses low-floor, high-ceiling task ideas and problem-solving habits from enrichment research. It does not label children, it is not an official GEP standard, and this program is not affiliated with MOE.

Ask for the reason every time: “How do you know?” A correct answer with no reason is only half done.

Answers are at the back. Do not hand the answer pages to children.

What each lesson practises

Lesson 1	Count with the Frog (Number Patterns)	8 + 12 + 6 questions
Lesson 2	Number Bond Inquiry (Number Bonds)	9 + 12 + 7 questions
Lesson 3	Let’s Plan the Fruit Party (Picture Graphs)	7 + 8 + 7 questions
Lesson 4	Pattern Inquiry (Repeating and Growing Patterns)	7 + 13 + 5 questions

Lesson 5	Mystery Number (Working Backwards)	8 + 10 + 6 questions
Lesson 6	Coin Combos (Money)	8 + 10 + 6 questions
Lesson 7	Shape Safari (2D Shapes)	7 + 9 + 6 questions
Lesson 8	Logic Lab (Logic and Sorting)	7 + 9 + 6 questions
Lesson 9	Measure It! (Length)	7 + 10 + 5 questions
Lesson 10	Puzzle Day (Mixed Puzzles)	6 + 9 + 4 questions

Lesson 1: Count with the Frog

Number Patterns • Week 1

CORE (low floor: start with objects, drawing or counting)

C1. A frog starts on 0 and hops 2 each time. Colour its landing places on a number strip up to 20 (or hop a counter). Write the numbers you coloured.

C2. Count in 5s. Fill the gaps: 5, 10, __, 20, __, 30

C3. 3, 6, 9, 12, __, __. Put a counter on 3 and hop to 6, then to 9. How many steps is each hop? Write the next two numbers.

C4. Write the next two numbers and the rule: 4, 7, 10, 13, __, __

C5. 50, 45, 40, 35, __, __. Are the numbers going up or down? Write the next two numbers.

C6. Box 1 has 2 stickers, box 2 has 4 stickers and box 3 has 6 stickers. Draw box 4 and box 5. How many stickers are in box 5?

C7. Count in 10s: 10, 20, 30, __, __. What is the rule?

C8. Mei counts 1, 3, 5, 7 … Will she say 10? Keep counting with her to find out.

THINK HARDER (use this lesson's idea in a new setting)

S1. Fill in the missing numbers: 3, 8, __, 18, __, 28

S2. Fill in the missing numbers: 40, __, 28, __, 16, 10

S3. Is 30 in the pattern 4, 8, 12, 16 … ? How do you know?

S4. Ali counts in 2s from 2 up to 50. Mei counts in 5s from 5 up to 50. Which numbers do both of them say? What do you notice?

S5. Ali counts from 3 in 4s: 3, 7, 11, … Will he say 39? Will he say 40?

S6. 5, 10, 15, 22, 25, 30. One number is wrong. Which number does not belong? What should it be?

S7. An add pattern has 11 as its 3rd number and 19 as its 5th number. What is the rule? What is the 1st number?

S8. Mei says: "If a pattern adds 5 each time, every number in it ends in 0 or 5." Is she right? Try a pattern that starts at 2 and one that starts at 3.

S9. A pattern goes 50, 44, 38, 32 … Will the pattern ever say 0? How do you know?

S10. Frog A jumps in 3s from 0. Frog B jumps in 4s from 0. Which numbers up to 40 do both frogs land on? (Do not count 0.)

S11. I am in the counting patterns of 3s and of 5s. I am more than 10 and less than 40. Which numbers can I be?

S12. Ali saves $2 in week 1, $5 in week 2, $8 in week 3 and so on. In which week does he save $29? How much does he save in week 12?

GO FURTHER (after the topic: extend, justify, make up your own)

G1. A pattern adds a number bigger than 0 each time, its 5th number is 20, and it starts at a whole number bigger than 0. How many different patterns can you make? List the first number and the rule for each.

G2. Frog A starts at 3 and jumps 4 each time. Frog B starts at 5 and jumps 3 each time. Which numbers up to 40 do both frogs land on?

G3. 30, 25, 21, 18, 16, __. What comes next? Describe how the gaps change.

G4. 1, 2, 4, 7, 11, 16 … What is the 8th number? Explain how the pattern grows.

G5. Ali counts up from 2 in 3s. Mei counts up from 1 in 4s. Which numbers up to 50 are in both patterns? What do you notice about the gaps between them?

G6. Make up your own two counting patterns: choose a start and a jump for each, so that they share some numbers up to 50. Write both patterns and ring the shared numbers. What do you notice about the gaps between the shared numbers?

Lesson 2: Number Bond Inquiry

Number Bonds • Week 1

CORE (low floor: start with objects, drawing or counting)

C1. You have 10 counters. 3 are red and the rest are yellow. How many are yellow? Write 10 = 3 + __

C2. A ten-frame has 6 counters in it. How many more fill it up? Write 10 = 6 + __

C3. A ten-frame has 8 counters in it. How many more fill it up? Write 10 = __ + 8

C4. A page has 20 stickers. 5 are stars and the rest are hearts. How many hearts? Write 20 = 5 + __

C5. A box holds 20 pencils. 12 are inside. How many more fit in? Write 20 = __ + 12

C6. 6 + 4 = 10. Count on from 16 on a number strip. What is 16 + 4?

C7. Two bags are the SAME. Bag A has 5 counters and 3 counters. Bag B has 4 counters and some more. Use counters to make the bags equal. How many more counters go in bag B?

C8. Ali has 12 stickers in his two hands. His left hand has 4 more than his right hand. Use counters: how many stickers are in each hand?

C9. Write three different pairs of numbers that make 10.

THINK HARDER (use this lesson's idea in a new setting)

S1. Mei says 7 + 5 = 8 + 4. Ben says 7 + 5 = 8 + 5. Who is right? Show how you know without adding both sides.

S2. Siti puts 10 stickers into two pockets. The left pocket must have MORE stickers than the right pocket, and each pocket has at least 1 sticker. How many ways can she do it? List them in order.

S3. 7 + 3 = 10. What is 8 + 4? What is 8 + 2? What happened to the total each time, and why?

S4. Find all the pairs of numbers from 1 to 11 that make 12. (3 + 9 and 9 + 3 count as the same pair.) How many pairs are there?

S5. 13 + 7 = 10 + __ = 4 + __

S6. A + A + 4 = 20, and the two As stand for the same number. What number is A?

S7. How many pairs make 20 if both numbers are from 5 to 15? (7 + 13 and 13 + 7 count once.)

S8. Which one does NOT belong: 6 + 4, 7 + 3, 8 + 3, 5 + 5? How do you know?

S9. True or false? Do not add everything: (a) 9 + 4 = 4 + 9 (b) 6 + 5 = 5 + 6 + 1 (c) 8 + 3 = 3 + 8

S10. The same number goes in both boxes: [ ] + [ ] = 18. What is the number? Then the same number goes in all three boxes: [ ] + [ ] + [ ] = 18. What is the number now?

S11. 8 + 2 = 10. Use it to help you fill in the gaps: 28 + __ = 30, 38 + __ = 40, 98 + __ = 100.

S12. Two numbers add up to 9. Can they be the SAME number? How do you know? Then find two numbers that add up to 9 where one is 3 more than the other.

GO FURTHER (after the topic: extend, justify, make up your own)

G1. Three different numbers from 3 to 7 add to 15. What can they be? Find both ways.

G2. Three different numbers from 1 to 7 add up to 10. Find every set of three.

G3. How many different pairs make 8 if the order matters (0 + 8 and 8 + 0 are different)? How many if the order does not matter?

G4. Ben says: "I can put 1 to 6 in the magic triangle so that every side adds to 13." Is he right? Explain how you know.

G5. Two circles and one square add up to 12. One circle and two squares add up to 9. What number is each shape? Explain how you found it.

G6. Ali has 20 stickers. He keeps 9 and shares the rest between Ben and Mei. Ben gets 3 more than Mei. How many does each get?

G7. Change the magic triangle: use the numbers 2, 3, 4, 5, 6 and 7 instead of 1 to 6. Can you make every side add to the same total? Try it. Which totals can you make?

Lesson 3: Let's Plan the Fruit Party

Picture Graphs • Week 2

CORE (low floor: start with objects, drawing or counting)

C1. Draw a picture graph with one dot for each child: Apple 7, Banana 5, Orange 9, Grape 4. Which fruit got the most votes?

C2. A picture graph shows: Apple 7, Banana 5, Orange 9, Grape 4 (1 picture = 1 child). Which fruit got the fewest votes?

C3. A picture graph shows: Apple 7, Banana 5, Orange 9, Grape 4 (1 picture = 1 child). How many more children chose orange than grape? (Line the dots up side by side.)

C4. A picture graph shows: Apple 7, Banana 5, Orange 9, Grape 4 (1 picture = 1 child). How many children voted altogether? (Point to each dot as you count.)

C5. Draw a picture graph for the pets: Dog 6, Cat 4, Fish 6, Bird 2 (1 picture = 1 child). Which two pets got the same number of votes?

C6. Pets: Dog 6, Cat 4, Fish 6, Bird 2. How many more children chose dog than bird?

C7. Ask 8 people in your family or group: "Cat or dog?" Make a mark for each vote. Which got more? By how many?

THINK HARDER (use this lesson's idea in a new setting)

S1. 1 picture = 2 children. Red has 3 pictures, blue has 5 and green has 4. How many children chose blue?

S2. 1 picture = 2 children. Red has 3 pictures, blue has 5 and green has 4. How many children voted altogether?

S3. 1 picture = 2 children. Six children chose pizza. How many pictures should we draw?

S4. 1 picture = 2 children. A half picture means 1 child. Apple has 3 pictures and a half. How many children chose apple?

S5. Each picture stands for 2 children. Apple has 3 pictures, banana has 5 and orange has 2. How many more children chose banana than orange?

S6. 25 children voted: apple 8, banana 6, grape 4, and the rest chose orange. Draw the orange row. Which fruit won?

S7. The key is missing. Apple has 4 pictures and 12 children chose apple. Banana has 3 pictures. How many chose banana?

S8. Ali says: "Blue has 2 more pictures than red, so blue got 2 more votes." Is he right if 1 picture = 2 children? What if 1 picture = 5 children?

GO FURTHER (after the topic: extend, justify, make up your own)

G1. A picture graph has 12 pictures for three fruits (1 picture = 2 children). Apple has the most pictures and no two fruits have the same number. How many pictures could apple have? List every graph you can find.

G2. 1 picture = 3 children. Apple has 4 pictures, banana 2 and orange 5. Then 7 more children join and all choose banana. Which fruit wins now, and by how many votes?

G3. 12 children voted for three fruits. No two fruits got the same number of votes. The most votes was 5. How many votes did each fruit get?

G4. Class A has 10 children: apple 6, orange 4. Class B has 20 children: apple 8, orange 12. Which fruit won in each class? Now put both classes together: which fruit wins, and by how many votes? Is that surprising?

G5. 24 children voted. Apple got 6 votes. Orange got 6 more votes than banana. How many votes did banana and orange get?

G6. 1 picture = 2 children. 20 children voted for three fruits. A has 3 pictures. B has 2 more pictures than A. How many children chose C?

G7. Make up your own: 15 children vote for three fruits. Draw a picture graph where banana wins by exactly 2 votes (banana has 2 more than the fruit in second place) and no two fruits are tied. Then try to find a second graph that works.

Lesson 4: Pattern Inquiry

Repeating and Growing Patterns • Week 2

CORE (low floor: start with objects, drawing or counting)

C1. The pattern is circle, square, triangle, circle, square, triangle, circle … What is the 10th one? (Keep writing the pattern until you get there.)

C2. The pattern is star, heart, star, heart, star, heart … What is the 9th one?

C3. The pattern is red, red, blue, red, red, blue, red … What is the 8th one?

C4. The pattern is A, B, C, D, A, B, C … What is the 11th one? (Write the letters in a row and number them.)

C5. The pattern is cat, dog, dog, cat, dog, dog, cat … What is the 12th one?

C6. The pattern is sun, moon, sun, moon, sun, moon … What is the 15th one?

C7. Build 1 triangle with 3 sticks. Add a second triangle next to it. How many sticks did you use? Now add a third triangle. How many sticks in all?

THINK HARDER (use this lesson's idea in a new setting)

S1. Pattern: circle, triangle, triangle, square, circle, triangle, triangle, square … What is the 14th shape?

S2. Same pattern: how many triangles are in the first 20 shapes?

S3. Red, blue, blue, red, blue, blue … What colour is the 20th bead?

S4. Squares in a row need 4, 7, 10, … sticks. With 20 sticks, how many squares can you make? How many sticks are left over?

S5. star, circle, circle, triangle, star, circle, circle, triangle … What is the 23rd shape?

S6. A, B, B, C, A, B, B, C, … How many Bs are in the first 18 letters?

S7. A pattern repeats every 5 shapes. Is the 25th shape the same as the 5th? Why?

S8. Pattern: circle, triangle, triangle, square, circle, triangle, triangle, square … In which places do the squares come? Write the first five.

S9. Red, blue, blue, red, blue, blue … How many blue beads are in the first 12 beads?

S10. Tiles: the 1st pattern has 3 tiles, the 2nd has 5, the 3rd has 7. How many tiles in the 10th pattern? Which pattern has 31 tiles?

S11. Triangles made of sticks in a row use 3, 5, 7 sticks. Ben has 25 sticks. How many triangles can he make? How many sticks are left over?

S12. Squares in a row use 4, 7, 10 … sticks. Ben has 50 sticks. How many squares can he make? How many sticks are left?

S13. A pattern repeats: star, moon, sun, star, moon, sun … What is the 100th shape? Explain how you know without drawing 100 shapes.

GO FURTHER (after the topic: extend, justify, make up your own)

G1. Squares in a row use 4, 7, 10 … sticks. Mei says she can make a row of squares using EXACTLY 30 sticks. Can she? What about exactly 31 sticks? Explain how you know.

G2. Pattern A repeats star, moon, sun. Pattern B repeats red, blue. Put them together: the 1st place is a red star, the 2nd a blue moon, the 3rd a red sun … What is the 20th place? When do we first see a red star again?

G3. Beads repeat in a pattern of 5: red, red, blue, blue, green. How many green beads are in the first 47 beads?

G4. Dots: 1, 3, 6, 10, 15 … What are the next two numbers? What is the 10th number?

G5. Make up your own repeating pattern with a core of 5 beads (use up to 3 colours). Ask a partner: what colour is the 33rd bead? Write the answer and how you know.

Lesson 5: Mystery Number

Working Backwards • Week 3

CORE (low floor: start with objects, drawing or counting)

C1. I think of a number. I add 3. I get 8. What is my number? (Hide counters under a cup and check.)

C2. I think of a number. I add 6. I get 14. What is my number?

C3. I think of a number. I add 5. I get 12. What is my number?

C4. I think of a number. I add 9. I get 17. What is my number?

C5. I think of a number. I take away 4. I get 6. What is my number?

C6. I think of a number. I take away 7. I get 3. What is my number?

C7. Ben is 4th in a queue. There are 3 children behind him. Draw the queue. How many children are in it?

C8. There are 20 children in a queue. Mei is 7th from the front. Draw it. How many children are behind her?

THINK HARDER (use this lesson's idea in a new setting)

S1. Ben had some marbles. He gave 5 to Mei, then won 3. Now he has 12. How many did he start with?

S2. Ben had some stickers. He gave away 6, got 4, then gave away 3. Now he has 10. How many did he have at first?

S3. A number machine adds the same number each time. IN 4 gives OUT 9. What is the rule? What is OUT when IN is 12? What is IN when OUT is 20?

S4. I think of a number. I add 4, then I take away 7. I get 8. What is my number?

S5. Tom is 5th from the front. There are 4 children between Tom and Sue, who is last. How many are in the queue?

S6. Mei had some sweets. Her mum gave her 6 more. Then she ate 4. Now she has 9. How many did she have at first?

S7. I think of a number. I take away 5, then I add 8. I get 14. What is my number?

S8. A machine adds 6, then takes away 2. IN 5 gives OUT 9. Can you make a one-step machine that does the same job? Check it with IN 10.

S9. I think of a number. I take away 4, then add 9, then take away 2. I get 20. What is my number?

S10. A machine gives OUT 7 for IN 3, OUT 9 for IN 4 and OUT 11 for IN 5. Is the machine "add 4"? How do you know? What does OUT do when IN goes up by 1?

GO FURTHER (after the topic: extend, justify, make up your own)

G1. Zoe is 6th from the front of a queue and 6th from the back. How many children are in the queue? What if a child is 10th from the front and 10th from the back? Can you find a quick way?

G2. Ali and Ben have 20 marbles together. Ali has 4 more than Ben. How many marbles does each have?

G3. I think of a number. I add 7, take away 3, add 5, then take away 8. I get 14. What is my number? Can you find a shortcut?

G4. A number machine has two steps: add 3, then add 4. IN 5 gives OUT 12. What goes IN to give OUT 20? Can you make a one-step machine that does the same job?

G5. A snail climbs 3 cm each day and slides back 1 cm each night. It starts at the bottom of a 9 cm wall. On which day does it reach the top?

G6. Make up a mystery number story with THREE steps whose answer is 10. Write the story. Show how a partner can work backwards to find 10, and how they can check by going forwards.

Lesson 6: Coin Combos

Money • Week 3

CORE (low floor: start with objects, drawing or counting)

C1. Pay exactly 15¢ using two coins from 5¢, 10¢, 20¢ and 50¢. (Use play coins.)

C2. Pay exactly 40¢ using two coins from 5¢, 10¢, 20¢ and 50¢.

C3. Pay exactly 55¢ using two coins from 5¢, 10¢, 20¢ and 50¢.

C4. How many different ways can you make 20¢ using only 10¢ and 5¢ coins? Use play coins and write each way.

C5. Ali has a 20¢ coin, a 10¢ coin and a 5¢ coin. How much money does he have?

C6. Mei has two 20¢ coins and one 5¢ coin. How much money does she have?

C7. Pay exactly 60¢ with the fewest coins from 5¢, 10¢, 20¢ and 50¢. (Use play coins.)

C8. Pay exactly 35¢ using 20¢, 10¢ and 5¢ coins. Use play coins and find a way with 3 coins.

THINK HARDER (use this lesson's idea in a new setting)

S1. Make 70¢ using exactly 4 coins from 5¢, 10¢, 20¢ and 50¢. Find every set.

S2. Sam has only 10¢ and 20¢ coins. He pays exactly 90¢ using 6 coins. How many of each coin does he use?

S3. What is the fewest number of coins to make 45¢?

S4. Find ALL the ways to make 30¢ using 20¢, 10¢ and 5¢ coins. How do you know you have found them all?

S5. How many ways can you make 40¢ with 20¢, 10¢ and 5¢ coins?

S6. Pay exactly $12 using exactly 3 notes or coins from $1, $2, $5 and $10. Find every way.

S7. What is the fewest number of coins for 95¢ (5¢, 10¢, 20¢, 50¢)? How do you know fewer is not possible?

S8. Ali pays 65¢ with the fewest coins he can (coins: 5¢, 10¢, 20¢, 50¢). Ben pays 65¢ with exactly 5 coins. How many more coins does Ben use than Ali?

S9. Ali buys a ruler for 35¢. He pays with a 50¢ coin. Which coins could he get back as change?

S10. A bag costs $34. Ali pays with a $50 note. How much change does he get?

GO FURTHER (after the topic: extend, justify, make up your own)

G1. You have one 5¢, one 10¢ and one 20¢ coin. How many different totals can you make using one, two or all three coins? List them.

G2. Which amounts in 5¢ steps (5¢, 10¢, 15¢ … 50¢) cannot be made using only 20¢ and 10¢ coins? What do you notice?

G3. Ali pays 55¢ with exactly 5 coins. He uses at least one 5¢, one 10¢ and one 20¢ coin. Which coins does he use?

G4. You have one 5¢, one 10¢, one 20¢ and one 50¢ coin. How many different amounts can you make using any of them? What if you also had a $1 coin (100¢)?

G5. A sweet costs 15¢ and a biscuit costs 20¢. Ben spends exactly 90¢ and buys some of each. How many sweets and biscuits does he buy?

G6. Make up your own coin puzzle that has exactly ONE answer (for example: "exactly 3 coins that make 35¢"). Write it, swap with a partner and check their answer by listing every possibility.

Lesson 7: Shape Safari

2D Shapes • Week 4

CORE (low floor: start with objects, drawing or counting)

C1. How many sides and corners does a triangle have? Draw one and count.

C2. How many sides and corners does a square have? Draw one and count.

C3. Draw a pentagon (a shape with 5 straight sides). How many corners does it have?

C4. Draw a hexagon (a shape with 6 straight sides). How many corners does it have?

C5. I have no corners and no straight sides. What shape am I?

C6. I have 3 straight sides and 3 corners. What shape am I?

C7. Fold a square piece of paper from corner to corner and cut along the fold. Which shapes do you get?

THINK HARDER (use this lesson's idea in a new setting)

S1. Two same-size triangles can make a square. Can they also make a bigger triangle? Draw how.

S2. How many squares are in a grid 3 squares long and 2 squares high?

S3. A square has both diagonals drawn (lines joining opposite corners). How many triangles can you find? Count the small ones first.

S4. Draw a shape with 4 straight sides that is NOT a square. Is it a rectangle? How do you know?

S5. Three squares are joined in a row to make a long rectangle. How many rectangles can you find altogether? (A square counts as a rectangle.)

S6. How many rectangles are in a grid 2 squares long and 2 squares high? (Squares count too.)

S7. Cut a square along both diagonals to get 4 same-size triangles. Can you put all 4 together to make a bigger triangle? Can you make a square? Draw them.

S8. How many squares are in a grid 3 squares long and 3 squares high?

S9. Ali cuts a rectangle along one line and gets two triangles. Where did he cut?

GO FURTHER (after the topic: extend, justify, make up your own)

G1. Join 4 same-size squares, whole sides touching. How many different shapes can you make? (Turns and flips are the same shape.) How do you know you have found them all?

G2. How many squares are in a grid 4 squares long and 2 squares high? What about 4 long and 3 high?

G3. A big triangle has 1 line from the top corner to the bottom. How many triangles? What about 2 lines? 3 lines?

G4. A big triangle has lines drawn from its top corner to the bottom. With 1 line there are 3 triangles, with 2 lines 6, with 3 lines 10. How many triangles with 4 lines? With 5 lines? Draw it to check.

G5. Draw a square and cut it into 4 equal squares. Then cut ONE of those small squares into 4 even smaller squares. How many squares can you find altogether? Count the small ones first, then the bigger ones.

G6. Draw your own triangle-counting puzzle: a big triangle with some lines inside. Count the triangles. Swap with a partner. Do you both get the same number? If not, find out who missed some.

Lesson 8: Logic Lab

Logic and Sorting • Week 4

CORE (low floor: start with objects, drawing or counting)

C1. Ali is taller than Ben. Ben is taller than Mei. Who is the tallest? Who is the shortest? (Draw three people.)

C2. The cat is heavier than the hen. The hen is heavier than the mouse. Who is the lightest?

C3. Red, blue and green ribbons. Red is longer than blue. Green is shorter than blue. Which ribbon is the longest?

C4. Ben stands in front of Mei. Mei stands in front of Sam. Who is in the middle?

C5. Odd numbers end in 1, 3, 5, 7 or 9. Which number is the odd one out: 2, 4, 6, 9, 10? Why?

C6. Odd numbers end in 1, 3, 5, 7 or 9. Even numbers end in 0, 2, 4, 6 or 8. Sort these numbers into odd and even: 3, 8, 5, 12, 7, 10.

C7. Use two loops of string. Put "red things" in one loop and "round things" in the other. Where does a red ball go?

THINK HARDER (use this lesson's idea in a new setting)

S1. In a race, Sam is ahead of Tia. Tia is ahead of Raj. Mei is ahead of Sam. Who is third?

S2. Ana, Bo and Cy each have a different toy: a ball, a kite or a yo-yo. Bo does not have the kite. Cy has the ball. Who has which toy?

S3. Numbers 1 to 12. Which do we say when counting in 2s? Which when counting in 3s? Which in both?

S4. Numbers 1 to 12. Which do we NOT say when counting in 2s or when counting in 3s?

S5. Ali, Ben, Cara and Dan each have a different pet: a cat, a dog, a fish or a rabbit. Cara has the dog or the fish. Cara does not have the dog. Dan does not have the cat or the rabbit. Ali does not have the cat. Who has which pet?

S6. Five children stand in a line. Ali is in the middle. Ben is just in front of Ali. Cara is last. Dan is not first. Where is Eve?

S7. Numbers 1 to 30: which do you say when counting in 2s AND when counting in 5s?

S8. Ali, Ben and Mei have a cat, a dog and a fish. Ali has no dog. Mei has no fish. Ben has neither a cat nor a fish. Who has which pet?

S9. Five friends run a race. D is first. E is last. B finishes directly behind A. C is not second. In what order do they finish?

GO FURTHER (after the topic: extend, justify, make up your own)

G1. Three boxes are labelled "Apples", "Oranges" and "Mixed". Every label is wrong. You may take one fruit from one box. Which box should you pick from, and why?

G2. A number between 20 and 40 is odd, and we say it when counting in 5s. It is also one less than a number we say when counting in 4s. What is it?

G3. Ali, Ben and Mei stand in a line. List every different order they can stand in. How many are there? How do you know you have them all? If Ben must NOT be first, how many orders are left?

G4. A secret number is between 10 and 30. It is even, and you say it when counting in 3s. It is 2 more than a number you say when counting in 5s. What is it?

G5. Numbers 1 to 20. Sort them in a two-loop picture with the loops "I say it counting in 2s" and "I say it counting in 3s". How many numbers go in each part? Then use the loops "counting in 2s" and "counting in 4s". What is different, and why?

G6. Make up your own "Who has which pet?" puzzle with three friends and three pets. Write clues so that there is only ONE answer. Swap with a partner and check that they find your answer.

Lesson 9: Measure It!

Length • Week 5

CORE (low floor: start with objects, drawing or counting)

C1. A pencil is 7 cubes long. A pen is 10 cubes long. How many cubes longer is the pen?

C2. A ribbon is 12 cubes long. You cut off 5 cubes. How long is the piece left?

C3. Ali's book is 9 cubes long and Mei's book is 6 cubes long. Whose is longer? By how much?

C4. Put in order from shortest: 11 cubes, 7 cubes, 9 cubes, 12 cubes.

C5. A ribbon lies from the 2 cm mark to the 7 cm mark on a ruler. How long is it? (Put a finger on 2 and hop to 7, counting each space.)

C6. A string is 9 cubes long. A rope is 4 cubes longer. How long is the rope?

C7. Line up cubes beside your shoe with no gaps. Guess first: how many cubes long is it? Then count.

THINK HARDER (use this lesson's idea in a new setting)

S1. A ribbon lies from the 3 cm mark to the 11 cm mark on a ruler. How long is it?

S2. A red ribbon lies from the 2 cm mark to the 9 cm mark. A blue ribbon lies from the 5 cm mark to the 11 cm mark. Which ribbon is longer? By how many cm?

S3. 1 clip is as long as 3 cubes. A pen is 12 cubes long. How many clips long is the pen? A ruler is 5 clips long. How many cubes long is the ruler?

S4. A pencil is 3 erasers long. A ruler is 2 pencils long. A desk is 4 rulers long. How many erasers long is the desk?

S5. A ribbon 15 cubes long is cut into 3 equal pieces. How long is each piece? How many cuts?

S6. A string is 6 paper clips long. 1 paper clip is as long as 2 cubes. Ben says the string is 12 cubes long. Mei says it is 8 cubes long. Who is right? How do you know?

S7. A pencil is 8 cm long. It starts at the 3 cm mark on a ruler. At which mark does it end?

S8. A rope is 8 cubes long. A string is 8 paper clips long. Do we know which is longer? Why not?

S9. A caterpillar is 9 cubes long. It grows 2 cubes each day. How long is it after 3 days?

S10. A rope is 8 cubes long. A string is 3 clips long. 1 clip is as long as 3 cubes. Which is longer, and by how many cubes?

GO FURTHER (after the topic: extend, justify, make up your own)

G1. A rope is 20 cubes long. Ali cuts it into pieces that are each 4 cubes long. How many pieces does he get? How many cuts does he make? Ali says, "There is always one cut fewer than pieces." Is

he right? Why?

G2. Ben lays a 12-cube stick end to end 3 times and has 5 cubes left over to reach the end of a desk. Mei measures the same desk with a 9-cube stick. How many sticks does she lay, and how many cubes are left over?

G3. A train has 3 carriages. Each is 6 cubes long. There is a gap of 1 cube between carriages. How long is the whole train? What if there are 5 carriages?

G4. A ribbon 30 cubes long is cut into three pieces. The first is 4 cubes longer than the second. The third is 4 cubes shorter than the second. How long is each piece?

G5. Make up your own measuring puzzle that uses TWO different units (for example clips and cubes). Write the puzzle and its answer. Give it to a partner and check their answer.

Lesson 10: Puzzle Day

Mixed Puzzles • Week 5

CORE (low floor: start with objects, drawing or counting)

C1. Count in 3s: 3, 6, 9, __, __

C2. 10 = 4 + __ and 20 = 12 + __

C3. Red 4, blue 6, green 3. How many more are blue than green?

C4. I think of a number. I add 4. I get 9. What is it?

C5. Make 15¢ with two coins from 5¢, 10¢ and 20¢.

C6. How many sides does a hexagon have? (Draw one if you like.)

THINK HARDER (use this lesson's idea in a new setting)

S1. A pattern starts at 4 and adds 6 each time. What is the 8th number? Is 50 in the pattern?

S2. Mei thinks of a number, adds 7, takes away 2 and gets 19. Ben does the same and gets 25. How much bigger is Ben's number?

S3. Ali has 5 coins worth 60¢. He has only 5¢, 10¢ and 20¢ coins. Find every possible set.

S4. Votes: Apple 6, Banana 9, Orange 7. Then 5 more children vote and all choose apple. Which fruit wins now? By how many votes?

S5. A ribbon 10 cubes long is cut into pieces that are 2 cubes or 3 cubes long, with nothing left over. List every way.

S6. Ben is in front of Mei. Sam is last. Ali is not first. Mei is in front of Ali. In what order do the four children stand?

S7. Ali has 7 coins. They are all 10¢ and 5¢ coins and are worth 50¢. How many of each coin does he have?

S8. 4, 9, 14, 19, ... Which number in this pattern is closest to 50?

S9. Mei gets 2 stickers from Ali, then gives 3 stickers back to Ali. Now Mei has 4 stickers. How many stickers did Mei have at first?

GO FURTHER (after the topic: extend, justify, make up your own)

G1. A secret number is between 20 and 60. We say it when we count in 3s and when we count in 5s. It is also one more than a number we say when we count in 4s. What is it?

G2. Triangles in a row use 3, 5, 7 ... sticks. Squares in a row use 4, 7, 10 ... sticks. Which numbers of sticks, up to 31, can make a row of triangles AND (with the same number of sticks) a row of squares, with none left over? What do you notice?

G3. Ali has some stickers. Ben has 4 more than Ali. Mei has 3 fewer than Ben. Together they have 29. How many does each have?

G4. Pick any puzzle from today (a puzzle card or a worksheet question). Change ONE thing about it (a number, a rule or a condition) to make a new puzzle. Solve your new puzzle and say what changed in the answer, and why.

Answer key (tutor only)

Lesson 1: Count with the Frog

C1. 2, 4, 6, 8, 10, 12, 14, 16, 18, 20.

C2. 15 and 25.

C3. Each hop is 3 steps. The next two numbers are 15 and 18.

C4. 16, 19. Add 3.

C5. Down. 30, 25. Take away 5.

C6. 10 stickers (2 more each time).

C7. 40, 50. Add 10.

C8. No. She says 9 and then 11, so she jumps over 10.

S1. 13 and 23. Add 5.

S2. 34 and 22. Take away 6 (use the two neighbours 16 and 10).

S3. No. The pattern goes 28, 32, so it skips 30.

S4. 10, 20, 30, 40 and 50. They all end in 0.

S5. Yes to 39, no to 40 (3, 7, 11, 15, 19, 23, 27, 31, 35, 39, 43: it lands on 39 and jumps over 40).

S6. 22 does not belong. It should be 20 (the pattern adds 5).

S7. Add 4: from the 3rd to the 5th number is two equal jumps that make 8, and 4 + 4 = 8. The 2nd number is 11 − 4 = 7 and the 1st is 7 − 4 = 3. Check: 3, 7, 11, 15, 19.

S8. No. It only works when the pattern starts at 0 or at a number that ends in 0 or 5. Starting at 2: 2, 7, 12, 17 … the numbers end in 2 or 7. Starting at 3: 3, 8, 13, 18 … they end in 3 or 8.

S9. No. Keep taking away 6: 50, 44, 38, 32, 26, 20, 14, 8, 2. The next step would go below 0, so it jumps over 0.

S10. 12, 24 and 36.

S11. 15 and 30.

S12. Week 10 ($29). Week 12: $35. The rule is add 3.

G1. 4 patterns: start 16 add 1; start 12 add 2; start 8 add 3; start 4 add 4. (Start 0 or a jump of 5 would need a first number of 0.)

G2. 11, 23 and 35.

G3. 15. The gaps shrink: take away 5, then 4, 3, 2 and now 1.

G4. 29. The gaps are 1, 2, 3, 4, 5, 6, 7, so the 7th is 22 and the 8th is 29.

G5. 5, 17, 29 and 41. Each one is 12 more than the one before.

G6. Open. Check by writing both lists and ringing the matches. Example: start 2 add 4 (2, 6, 10 …) and start 3 add 3 (3, 6, 9 …) share 6, 18, 30 and 42, with a gap of 12 each time.

Lesson 2: Number Bond Inquiry

C1. 7

C2. 4

C3. 2

C4. 15

C5. 8

C6. 20

C7. 4 (5 + 3 = 8 and 4 + 4 = 8).

C8. Left 8, right 4.

C9. Open. For example 1 + 9, 2 + 8, 3 + 7, 4 + 6, 5 + 5 (any three).

S1. Mei. 8 is 1 more than 7 and 4 is 1 less than 5, so the total stays the same (both sides make 12). Ben made the first number 1 bigger but did not make the second number smaller, so his right side is 1 too big (13).

S2. 4 ways: 6 and 4, 7 and 3, 8 and 2, 9 and 1. Going down one at a time from 6 and 4 shows none are missed (5 and 5 is a tie, and 10 and 0 leaves a pocket empty).

S3. 12 and 10. 8 + 4: both numbers went up by 1, so the total went up by 2. 8 + 2: one number went up by 1 and the other went down by 1, so the total stayed the same.

S4. 6 pairs: 1 + 11, 2 + 10, 3 + 9, 4 + 8, 5 + 7, 6 + 6. Going down one number at a time shows none are missed.

S5. 10 and 16 (all three are 20).

S6. 8 (20 − 4 = 16, and 8 + 8 = 16).

S7. 6 pairs: 5 + 15, 6 + 14, 7 + 13, 8 + 12, 9 + 11, 10 + 10.

S8. 8 + 3. It makes 11 and the others all make 10.

S9. (a) True (b) False: the right side is 1 more (c) True. Turning a sum round does not change the total.

S10. 9 (9 + 9 = 18), and 6 (6 + 6 + 6 = 18).

S11. 2 each time. The ones stay the same (8 + 2 = 10), so each answer is 2 away from the next ten.

S12. No. 4 + 4 = 8 and 5 + 5 = 10, so no whole number added to itself makes 9. The numbers are 3 and 6 (3 + 6 = 9 and 6 is 3 more than 3).

G1. 3, 5, 7 and 4, 5, 6.

G2. 1, 2, 7; 1, 3, 6; 1, 4, 5; 2, 3, 5.

G3. 9 pairs when the order matters; 5 pairs when it does not (0+8, 1+7, 2+6, 3+5, 4+4).

G4. No. The biggest side total is 12: put the biggest numbers 4, 5 and 6 in the corners (each corner is counted in two sides) and the sides are 4 + 3 + 5, 5 + 1 + 6 and 6 + 2 + 4, all 12. Any other corners give less, so 13 is not possible. (Possible totals: 9, 10, 11, 12.)

G5. Circle 5, square 2 (2 × 5 + 2 = 12 and 5 + 2 × 2 = 9). One way to see it: the two totals differ by 3, and the only change is one circle swapped for one square, so a circle is 3 more than a square. Then 3 squares and 6 more make 12, so each square is 2.

G6. Ben 7, Mei 4. (20 – 9 = 11 are shared; take away Ben's extra 3: 8, share 4 and 4; Ben has 4 + 3 = 7.)

G7. Open. Yes, it still works. Totals 12, 13, 14 and 15 are possible. Example for 12: corners 2, 3, 4 with 7 between 2 and 3, 5 between 3 and 4 and 6 between 4 and 2.

Lesson 3: Let's Plan the Fruit Party

C1. Orange

C2. Grape

C3. 5 (9 – 4)

C4. 25 (7 + 5 + 9 + 4)

C5. Dog and fish

C6. 4 (6 – 2)

C7. Open. Check the two totals add up to the number of people asked.

S1. 10

S2. 24 (12 pictures, each worth 2: count in 2s)

S3. 3

S4. 7

S5. 6 more. Banana 2, 4, 6, 8, 10 children; orange 2, 4 children; 10 – 4 = 6 (or: 3 extra pictures, count in 2s: 2, 4, 6).

S6. Orange 7 (8 + 6 + 4 = 18, count on from 18 to 25: 7). Apple wins with 8.

S7. 9. Count in 3s: 3, 6, 9, 12 is 4 pictures, so 1 picture = 3 children. Banana: 3, 6, 9.

S8. No. Each extra picture is worth 2 children, so blue got 4 more votes (2 pictures, each worth 2). If 1 picture = 5 children, blue got 10 more votes.

G1. Apple could have 5, 6, 7, 8 or 9 pictures. There are 7 graphs (apple, second, third): 9, 2, 1; 8, 3, 1; 7, 4, 1; 6, 5, 1; 7, 3, 2; 6, 4, 2; 5, 4, 3.

G2. Orange 15, banana 13, apple 12. Orange wins by 2.

G3. 5, 4 and 3.

G4. Apple wins in Class A, orange wins in Class B. Together apple 6 + 8 = 14 and orange 4 + 12 = 16, so orange wins by 2. Class A alone would have told us the wrong fruit for the group: a small sample can mislead.

G5. Banana 6 and orange 12. (24 – 6 = 18 for banana and orange; take away the extra 6: 12, share: 6 each; give orange the 6 back.)

G6. 4 children (20 ÷ 2 = 10 pictures in all, counting in 2s to 20; A 3 + B 5 = 8, so C has 2 pictures = 4 children).

G7. Open. Check: total 15, banana is ahead of second place by exactly 2, no ties. There are exactly two graphs: banana 7, apple 5, grape 3; or banana 8, apple 6, grape 1.

Lesson 4: Pattern Inquiry

C1. circle

C2. star

C3. red

C4. C

C5. dog

C6. sun

C7. 5 sticks for 2 triangles and 7 sticks for 3 triangles (each new triangle shares a side).

S1. triangle

S2. 10. The core circle, triangle, triangle, square has 2 triangles. Count the cores in 4s: 4, 8, 12, 16, 20, so there are 5 cores. Count the triangles in 2s: 2, 4, 6, 8, 10.

S3. blue

S4. 6 squares and 1 stick left over. The table goes 4, 7, 10, 13, 16, 19; 7 squares would need 22, which is too many; 20 − 19 = 1.

S5. circle. The core has 4 shapes, so count in 4s: 4, 8, 12, 16, 20. The 20th is a triangle, the end of a core; 21st star, 22nd circle, 23rd circle.

S6. 9 Bs. Each core A, B, B, C has 2 Bs. 16 letters are 4 cores: 2, 4, 6, 8 Bs. The 17th is A and the 18th is B: 9.

S7. Yes. Count in 5s: 5, 10, 15, 20, 25. Each of these ends the core, so the 25th is the last shape of the core, like the 5th.

S8. 4, 8, 12, 16, 20

S9. 8

S10. 21 tiles in the 10th. 31 tiles is the 15th.

S11. 12 triangles use 25 sticks exactly. Keep adding 2: 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25. None left over.

S12. 16 squares use 49 sticks. 1 stick is left.

S13. Star. Every place we say when counting in 3s ends the core with a sun. We say 99 when counting in 3s, so the 99th is a sun and the 100th starts again: star.

G1. No for 30: the stick numbers go 4, 7, 10, 13, 16, 19, 22, 25, 28, 31 and skip 30 (each number is 1 more than a number we say counting in 3s, and 29 is not one). Yes for 31: that is 10 squares.

G2. The 20th place is a blue moon (20 is 2 after 18, so moon; 20 is even, so blue). A red star comes back at the 7th place, then the 13th: every 6 places (3 and 2 meet at 6).

G3. 9 green beads. Green is the 5th bead of each core, so greens are at 5, 10, 15 … 45: count in 5s, that is 9 numbers. Beads 46 and 47 are red.

G4. 21 and 28. The 10th is 55 (add 2, 3, 4, … each time).

G5. Open. Check: 30 ends a core of 5 (count in 5s), so the 33rd bead is the 3rd bead of the core. Example: red, red, blue, blue, green gives blue.

Lesson 5: Mystery Number

C1. 5

C2. 8

C3. 7

C4. 8

C5. 10

C6. 10

C7. 7

C8. 13 (20 − 7)

S1. 14 (12 − 3 + 5)

S2. 15 (work backwards: 10 + 3 = 13, 13 − 4 = 9, 9 + 6 = 15; check 15 − 6 + 4 − 3 = 10)

S3. Add 5. OUT is 17 when IN is 12. IN is 15 when OUT is 20.

S4. 11

S5. 10 (Tom is 5th, then 4 children, then Sue: 5 + 4 + 1).

S6. 7 (work backwards: 9 + 4 = 13, 13 − 6 = 7; check 7 + 6 − 4 = 9).

S7. 11 (undo in reverse order: 14 − 8 = 6, 6 + 5 = 11).

S8. Yes: add 4. 5 + 4 = 9 and 10 + 4 = 14 (10 + 6 − 2 = 14 too).

S9. 17. Work backwards: 20 + 2 = 22, then 22 − 9 = 13, then 13 + 4 = 17. Check: 17 − 4 = 13, 13 + 9 = 22, 22 − 2 = 20.

S10. No. 4 + 4 is 8, not 9. When IN goes up by 1, OUT goes up by 2 (7, 9, 11), so the machine is not adding the same number each time.

G1. 11 and 19. Zoe is counted twice if we add 6 + 6, so take away 1. Quick way: double the place number and take away 1 (6 + 6 − 1 = 11; 10 + 10 − 1 = 19).

G2. Ben 8 and Ali 12. (Take away the extra 4: 16 left, share: 8 and 8, give Ali the 4 back: 12.)

G3. 13. The steps are +7, −3, +5, −8: together they are +1, so the number is 14 − 1.

G4. 13 (20 − 4 = 16, 16 − 3 = 13; check 13 + 3 + 4 = 20). A one-step machine that adds 7 does the same job.

G5. Day 4 (3, 5, 7, then 9 on the fourth day before it can slide back).

G6. Open. Check that the story gives 10 when done forwards and that the partner undoes the steps in the reverse order. Example: I think of a number, add 5, take away 3, add 4, and I get 16 (10 + 5 = 15, 15 – 3 = 12, 12 + 4 = 16).

Lesson 6: Coin Combos

C1. 5¢ + 10¢

C2. 20¢ + 20¢

C3. 5¢ + 50¢

C4. 3 ways: 10 + 10; 10 + 5 + 5; 5 + 5 + 5 + 5.

C5. 35¢

C6. 45¢

C7. 50¢ + 10¢ (2 coins)

C8. 20¢ + 10¢ + 5¢ (3 coins)

S1. 2 sets: 50¢ + 10¢ + 5¢ + 5¢, and 20¢ + 20¢ + 20¢ + 10¢.

S2. Three 10¢ and three 20¢ (30 + 60 = 90).

S3. 3 coins (20¢ + 20¢ + 5¢)

S4. 6 ways: 20+10; 20+5+5; 10+10+10; 10+10+5+5; 10+5+5+5+5; six 5¢. Start with the biggest coin and work down.

S5. 9 ways (two 20¢: 1 way; one 20¢: 3 ways; no 20¢: 5 ways).

S6. 2 ways: $10 + $1 + $1, and $5 + $5 + $2.

S7. 4 coins: 50 + 20 + 20 + 5. Three coins cannot work: with a 50¢ coin you need 45¢ from two coins (no pair makes 45¢), and without it three coins make at most 60¢.

S8. Ali uses 3 coins (50¢ + 10¢ + 5¢). Ben uses 20¢ + 20¢ + 10¢ + 10¢ + 5¢, which is the only way with 5 coins. Ben uses 2 more coins.

S9. 15¢: 10¢ + 5¢ (or three 5¢ coins).

S10. $16. Count on from $34: $36, $40, $50. That is $2 + $4 + $10 = $16.

G1. 7 totals: 5, 10, 15, 20, 25, 30, 35.

G2. 5¢, 15¢, 25¢, 35¢ and 45¢. (Every amount that does not end in 0.)

G3. 20¢ + 10¢ + 10¢ + 10¢ + 5¢ (the only way).

G4. 15 different amounts: with 3 coins there are 7, and the 50¢ coin doubles that and adds one more. With a $1 coin as well there are 31: each new coin doubles the amounts and adds one more.

G5. 2 sweets and 3 biscuits.

G6. Open. A good puzzle has exactly one way. Example: 35¢ with exactly 3 coins from 5¢, 10¢, 20¢ and 50¢ has only 20¢ + 10¢ + 5¢. Check by listing every set of 3 coins.

Lesson 7: Shape Safari

C1. 3 sides and 3 corners

C2. 4 sides and 4 corners

C3. 5 corners

C4. 6 corners

C5. A circle

C6. A triangle

C7. 2 triangles of the same size

S1. Yes. Join them along a short side (the two equal short sides touching) to make one bigger triangle. Joining along the long side makes a square instead.

S2. 8 (6 small squares + 2 squares made of 4 small ones)

S3. 8 (4 small + 4 made of two small ones)

S4. Open. A long rectangle has 4 square corners, so it is a rectangle. A slanted shape with 4 sides does not have square corners, so it is not a rectangle.

S5. 6 (3 single squares, 2 pairs and 1 whole row).

S6. 9: 4 small squares, 2 rectangles 2 long lying down, 2 rectangles 2 long standing up, and the whole grid.

S7. Yes to both. The 4 triangles make the original square again (corners meeting in the middle), and they also fit together into one big triangle whose long side is twice as long as a small triangle's long side (two along the bottom and two on top).

S8. 14 (9 small + 4 made of 4 small ones + 1 made of all 9)

S9. From one corner to the opposite corner.

G1. 5 shapes: a line of 4, a 2 by 2 square, an L, a T and a zigzag. Build them in order: start with a line of 4, then move one square at a time and check each new one is not a turned or flipped copy.

G2. 11 (8 small + 3 of size 2 by 2) and 20 (12 small + 6 of size 2 by 2 + 2 of size 3 by 3).

G3. 3, 6 and 10 triangles.

G4. 15 triangles with 4 lines and 21 with 5 lines. The number added grows by 1 each time: 3, 6, 10, 15, 21 (+3, +4, +5, +6). Check by drawing: count the small triangles first, then the bigger ones made of 2, 3 … small ones.

G5. 9 (1 big, 4 medium, 4 tiny).

G6. Open. Check by counting small triangles first, then bigger ones made of 2, 3 … small ones. Example: one line from the top corner to the bottom gives 3 triangles; two lines give 6.

Lesson 8: Logic Lab

C1. Ali is tallest; Mei is shortest.

C2. The mouse

C3. Red

C4. Mei

C5. 9. It is the only odd number.

C6. Odd: 3, 5, 7. Even: 8, 12, 10.

C7. In the middle, where the two loops overlap (it is red AND round).

S1. Tia (order: Mei, Sam, Tia, Raj).

S2. Cy: ball. Bo: yo-yo. Ana: kite.

S3. Counting in 2s: 2, 4, 6, 8, 10, 12. Counting in 3s: 3, 6, 9, 12. Both: 6 and 12.

S4. 1, 5, 7 and 11

S5. Cara fish (dog or fish, not dog). Dan: not cat, not rabbit, fish taken, so dog. Ali: not cat, so rabbit. Ben: cat.

S6. Eve is first. Ali is 3rd (the middle of 5), Ben 2nd, Cara 5th; 1st and 4th are left for Dan and Eve, and Dan is not first, so Dan 4th and Eve 1st: Eve, Ben, Ali, Dan, Cara.

S7. 10, 20 and 30.

S8. Ben: dog. Mei: cat. Ali: fish.

S9. D, A, B, C, E. (A and B are side by side in places 2-3 or 3-4; C is not second, so A is 2nd and B is 3rd.)

G1. Pick from the box labelled "Mixed". Its label is wrong, so it holds only one kind of fruit. If you take out an apple, that box is Apples. The box labelled "Oranges" cannot be oranges, and apples are taken, so it is Mixed. The box labelled "Apples" must be Oranges.

G2. 35. The odd ones are 25 and 35. 25 + 1 = 26: not said counting in 4s. 35 + 1 = 36: yes.

G3. 6 orders: ABM, AMB, BAM, BMA, MAB, MBA (Ali, Ben, Mei). Fix who is first, then list the two ways for the other two. If Ben is not first: 4 orders.

G4. 12.

G5. 2s and 3s: 2s only 7, 3s only 3, both 3 (6, 12, 18), neither 7. 2s and 4s: 2s only 5, 4s only 0, both 5 (4, 8, 12, 16, 20), neither 10. Every number we say counting in 4s we also say counting in 2s, so the 4s loop sits completely inside the 2s loop.

G6. Open. Check by trying all 6 ways to hand out the pets. Example: Ana, Bo and Cy have a cat, a dog and a fish. Ana does not have the cat. Bo does not have the dog or the fish. Cy does not have the fish. Answer: Ana fish, Bo cat, Cy dog.

Lesson 9: Measure It!

C1. 3

C2. 7 cubes

C3. Ali's, by 3 cubes.

C4. 7, 9, 11, 12

C5. 5 cm (count the spaces: 7 − 2)

C6. 13 cubes

C7. Open. Check there are no gaps or overlaps.

S1. 8 cm (count the spaces from 3 to 11, or 11 − 3)

S2. Red is longer: red 7 cm, blue 6 cm, so red is 1 cm longer.

S3. 4 clips. 15 cubes.

S4. 24 erasers (a ruler is 3 + 3 = 6 erasers; a desk is 6 + 6 + 6 + 6 = 24)

S5. 5 cubes each (5 + 5 + 5 = 15) and 2 cuts: 3 pieces need only 2 cuts.

S6. Ben. Six clips of 2 cubes each: 2 + 2 + 2 + 2 + 2 + 2 = 12 cubes.

S7. The 11 cm mark (3 + 8 = 11).

S8. No. A cube and a paper clip are different lengths, so 8 of each are not the same.

S9. 15 cubes

S10. The string. 3 clips = 3 × 3 = 9 cubes, which is 1 cube longer than the rope.

G1. 5 pieces and 4 cuts. Yes: the last piece needs no cut, so cuts are always 1 fewer than pieces.

G2. The desk is 12 + 12 + 12 + 5 = 41 cubes long. Mei lays her 9-cube stick 4 times (36 cubes) and has 5 cubes left over.

G3. 20 cubes (18 + 2 gaps). With 5 carriages: 30 + 4 gaps = 34 cubes. There is always 1 gap fewer than carriages.

G4. 14, 10 and 6 cubes.

G5. Open. Check that the link between the units is stated and the answer works. Example: 1 clip is as long as 4 cubes. A snake is 5 clips long. How many cubes long is it? Answer: 20 cubes.

Lesson 10: Puzzle Day

C1. 12, 15

C2. 6 and 8

C3. 3

C4. 5

C5. 10¢ + 5¢

C6. 6

S1. 46 (4, 10, 16, 22, 28, 34, 40, 46). No: the pattern goes 46, 52, so it skips 50.

S2. 6 bigger. Work backwards: Mei 19 + 2 = 21, 21 − 7 = 14; Ben 25 + 2 = 27, 27 − 7 = 20; 20 − 14 = 6. (Shortcut: both did the same steps, so the answers differ by the same amount: 25 − 19 = 6.)

S3. Two sets: 20 + 10 + 10 + 10 + 10, and 20 + 20 + 10 + 5 + 5. Make a table by number of 20¢ coins: one 20¢ needs 40¢ from 4 coins (four 10¢); two 20¢ need 20¢ from 3 coins (10 + 5 + 5); three 20¢ need 0¢ from 2 coins (impossible); no 20¢: 5 coins make at most 50¢.

S4. Apple wins with 11, which is 2 more than banana (9).

S5. Two ways: five pieces of 2 cubes; or two pieces of 3 cubes and two pieces of 2 cubes (3 + 3 + 2 + 2 = 10). Three pieces of 3 leave 1 over, and one piece of 3 leaves 7, which 2s cannot fill.

S6. Ben, Mei, Ali, Sam.

S7. Three 10¢ and four 5¢ (30 + 20 = 50).

S8. 49 (list it: 24, 29, 34, 39, 44, 49, 54; 49 is 1 away, 54 is 4 away).

S9. 5. Work backwards: 4 + 3 = 7, then 7 − 2 = 5. Check: 5 + 2 = 7, 7 − 3 = 4.

G1. 45. Numbers in both the 3s and the 5s counts: 30, 45, 60. 30 − 1 = 29 and 60 − 1 = 59 are not said counting in 4s; 45 − 1 = 44 is.

G2. 7, 13, 19, 25 and 31. They go up by 6 (the 2 and the 3 from the two patterns meet every 6). For example 7 sticks make 3 triangles or 2 squares.

G3. Ali 8, Ben 12, Mei 9.

G4. Open. Check that the new puzzle can be solved and that the child explains the link between the change and the new answer. Example: change "Make 45¢ with exactly 3 coins" (20¢ + 20¢ + 5¢) to "Make 45¢ with exactly 4 coins": now the only way is 20¢ + 10¢ + 10¢ + 5¢.